

AYMANE AIT DADS

Software Engineer Intern - Observability & Monitoring | Python · Linux · Data Pipeline Engineering
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PROFESSIONAL SUMMARY

Built an end-to-end automated data pipeline on 100K+ fiber-optic network records at Orange Maroc, reducing manual analysis time by ~60% and delivering actionable system-health outputs directly to a network optimization team. Ranked top 10% on a \$106K+ Kaggle leaderboard by engineering training infrastructure for a 30B-parameter model on novel GPU hardware, resolving 5 distinct low-level CUDA/Triton incompatibilities through systematic ablation. Proficient in Python, Linux, Git, and data visualization (Power BI, Matplotlib), with hands-on experience in automated feature engineering, anomaly detection, and performance monitoring pipelines. Available for internship in Europe with no work-authorization restrictions; based in Sophia Antipolis, France.

CORE COMPETENCIES

Observability & Monitoring | Python & Linux Proficiency | Performance Profiling | Anomaly Detection | Data Pipeline Engineering | Git Version Control | Dashboard & Visualization | System Performance Analysis

WORK EXPERIENCE

Orange Maroc Jul 2024 - Aug 2024
Data Science Intern - Casablanca, Morocco

- Engineered an **end-to-end automated pipeline** processing 100K+ fiber-optic network records (upload speed, latency, jitter) to classify node-level network quality across thousands of infrastructure nodes.
- Developed a clustering pipeline mapping customer performance profiles to **5 commercial service tiers**, with output consumed directly by the network optimization team for maintenance decisions.
- Reduced **manual analysis time by ~60%** through automated feature engineering, cross-validated model evaluation, and Random Forest classification integrated with K-Means segmentation.
- Delivered stakeholder presentations and **Power BI dashboards** translating model outputs into maintenance recommendations, bridging technical results and operational decision-making.

PROJECTS

Anomalous Sound Detection - Industrial Equipment EURECOM · 2025 · AUC 0.80 Unsupervised Fault Detection

- Designed an unsupervised fault-detection system for slide-rail machinery using a Transformer autoencoder on Mel spectrograms with SpecAugment, achieving **AUC 0.80** with zero labeled anomaly examples.
- Built the full monitoring pipeline - feature extraction, model evaluation, and anomaly scoring - directly analogous to observability tooling that detects trends and regressions across system data streams.

PyTorch · Scikit-learn · NumPy · Matplotlib

NVIDIA Nemotron Reasoning Challenge Kaggle Competition · In Progress June 2026 · Top 10% Public Leaderboard

- Achieved **top 10% public leaderboard score (0.80/1.0)** in a \$106K+ prize-pool competition by fine-tuning a 30B Mamba-Transformer MoE model with LoRA (r=32, ~888M trainable parameters) via SFTTrainer on 7,828 chain-of-thought examples.
- Resolved **5 Blackwell GPU (sm_120) incompatibilities** - including OOM patching, Mamba CUDA kernel disabling, Triton ptxas permission fixes, and a PyTorch RMSNorm fallback - through 20+ single-variable ablations confirming strict fix-ordering requirements.

PyTorch · Unsloth · PEFT · Hugging Face Transformers · Triton

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Cloud Computing, Computer Vision, NLP, Statistics, Image Security, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya 2021 - 2023

SKILLS

Infrastructure & Observability: Python | Linux | Git | Bash | Docker (basic) | PostgreSQL | REST APIs
Data Engineering & Visualization: Pandas | NumPy | Power BI | Matplotlib | Seaborn | Feature Engineering | EDA
ML & Systems: PyTorch | Scikit-learn | Hugging Face Transformers | CUDA Debugging | FP16 Mixed Precision | Anomaly Detection